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ORIGINAL RESEARCH

The association between sociodemographic and clinical factors and outcomes among inpatients with community-acquired pneumonia

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Abstract

Background: Community-acquired pneumonia (CAP) is a leading cause of hospitalization and mortality. The influence of sociodemographic characteristics on adverse outcomes in adults hospitalized with CAP remains poorly understood, particularly for patients younger than 65.

Objectives: To evaluate associations between baseline sociodemographic characteristics and a composite outcome of intensive care unit (ICU) admission or in-hospital mortality, and whether these differed across age strata (18–39, 40–64, and ≥65 years).

Methods: We conducted a multicentre, retrospective, cohort study of adults ≥18 years hospitalized with CAP between 2015 and 2024 in Ontario. Using generalized linear mixed models stratified by age, we evaluated the adjusted effects of clinical (i.e., comorbidity and illness severity) and sociodemographic characteristics (i.e., age, sex, four dimensions of neighborhood-level marginalization) with the composite outcome. We compared the contribution to model fit of these variable groups using nested likelihood ratio tests.

Results: Among 84,356 hospitalizations, males had higher adjusted odds of the composite outcome across all age groups (adjusted odds ratio [aOR] range: 1.25–1.48), while residence in areas with higher concentrations of racialized/newcomer populations was associated with lower odds of the composite outcome (aOR range: 0.89–0.96). Nested likelihood ratio tests showed that clinical characteristics contributed substantially more to outcome variation than sociodemographic factors (χ^2 1682 vs. 145 in the ≥65 stratum).

Peter J. Gill and Sanjay Mahant contributed equally to this manuscript.

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Conclusions: Sociodemographic and clinical characteristics were associated with severe CAP outcomes, but clinical factors accounted for a greater proportion of outcome variation. Incorporating sociodemographic information, including sex, age, and marginalization measures, into early assessment may improve identification of patients at higher risk.

INTRODUCTION

Community-acquired pneumonia (CAP) is a leading cause of morbidity and mortality in adults worldwide.[1,2] In the United States, the annual incidence of hospitalization for CAP is approximately 650 hospitalizations per 100,000 adults each year, resulting in more than 1.5 million unique admissions annually.[3–5] The disease burden of hospitalized CAP among adults is substantial, with intensive care unit (ICU) admissions for CAP hospitalizations ranging from 6% in England[6] to 23% in the United States,[7] while mortality during hospitalization is between 7% and 10.6%.[4,6,8–10] There is also substantial economic impact, with costs estimated to be $11,148 USD for uncomplicated cases and $51,219 USD for complicated cases per inpatient episode.[5,11]

Several risk factors have been associated with poorer prognosis among hospitalized patients with CAP, including advanced age (> 65 years), male sex, abnormal vital signs, and comorbid conditions such as chronic obstructive pulmonary disease (COPD), chronic heart disease, diabetes, and cerebrovascular disease.[12–16] However, the influence of socioeconomic position and marginalization on CAP outcomes has been less well characterized. Existing studies suggest a trend toward higher marginalization being linked with more severe outcomes.[17–19] Understanding how baseline sociodemographic factors, specifically sex and marginalization, relate to poor outcomes for hospitalized patients with CAP may facilitate early risk stratification and improve clinical decision-making. Furthermore, whether these associations differ across age groups, particularly among younger adults, remains to be studied.

To address these gaps, we conducted a large multicenter retrospective cohort study of hospitalized adults with CAP to evaluate the association between baseline sociodemographic characteristics and adverse CAP outcomes, including ICU admission and in-hospital mortality, and to determine whether these associations were consistent across age strata (18–39, 40–64, and ≥65 years). We also characterized processes of care in the clinical management of CAP, including medical interventions, antibiotic use, laboratory tests, and radiology orders, by these age strata.

METHODS

Study design, setting, and data sources

We conducted a multicenter retrospective cohort study across 26 hospitals in Ontario, Canada, from April 1, 2015 to March 30, 2024, participating in the General Medicine Inpatient Initiative (GEMINI) hospital research network.[20] GEMINI includes administrative data submitted to the Canadian Institute for Health Information (CIHI) for the Discharge Abstract Database and National Ambulatory Care Reporting System[21] and clinical data from hospital-specific electronic medical record systems including demographics, diagnoses, interventions, laboratory and radiology results. Data sources, linkage, and quality checks are described in detail elsewhere.[20,22,23] Research ethics approval was obtained from all participating GEMINI hospitals.

Study sample

To identify patients (≥ 18 years) with CAP, we included all encounters with the most responsible discharge diagnosis of "pneumonia" based on the enhanced Canadian version of the *International Classification of Diseases and Related Health Problems, 10th Revision* (ICD-10-CA) codes J10-J18.[24–27] We also included encounters where pneumonia (J10-J18) was a secondary or comorbid diagnosis with a most responsible discharge diagnosis of chronic obstructive pulmonary disease (COPD, J41–J44), human immunodeficiency virus (HIV) (B20, B24, and Z21), or coronavirus disease 2019 (COVID-19) (U07).[28,29] This approach has been validated to have a sensitivity of 98% and specificity of 97% for CAP cases.[30] We also included encounters where aspiration pneumonia (ICD-10-CA code J69.0) was the most responsible diagnosis, as it is often difficult to distinguish CAP from aspiration pneumonia. The complete list of most responsible diagnoses is shown in Supporting Information S1: Table 1.

To increase the specificity and exclude hospital-acquired pneumonia, we only included encounters with at least one intravenous, oral, or enteral antibiotic order in the first 48 h of hospital admission.[31,32] Additionally, we excluded encounters with at least one order for an antifungal agent within the first 48 h of hospital admission, as these cases likely represent pneumonia due to fungal pathogens or pneumonia in an immunocompromised host, which would introduce heterogeneity to our cohort.

Baseline characteristics

Baseline patient characteristics were grouped into two domains for analysis: clinical characteristics (comorbidity burden and severity of illness) and sociodemographic characteristics (age, sex, and neighborhood-level marginalization). Clinical characteristics included

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overall level of comorbidity, estimated using the Charlson Comorbidity Index (CCI) score on admission (range 0–24, with higher scores indicating greater comorbidity), and the presence of each category of Charlson comorbidity.[33–35] Severity of illness at admission was estimated using the modified version of the Laboratory-based Acute Physiology Score (mLAPS) (range 0–231, with higher values indicating greater illness severity),[36,37] which performs similarly to the CURB-65 score (Confusion, Urea ≥ 42.8 mg/dL, Respiratory rate ≥ 30/min, Blood pressure systolic < 90 mmHg or diastolic ≤ 60 mmHg, and age ≥ 65 criteria) in CAP.[24]

Sociodemographic characteristics included age and sex at the patient level, and neighborhood-level marginalization measured using the Ontario Marginalization Index (ON-Marg). ON-Marg, an externally developed and validated area-based index produced by Public Health Ontario,[38] comprises four dimensions: (i) material resources, (ii) household and dwellings characteristics, (iii) racialized and newcomer populations, and (iv) age and labor force composition.[39] The dimensions were derived by the index developers using principal component factor analysis of 18 census-based indicators that were selected *a priori* from a theoretical framework of marginalization[40] and have been replicated across successive Canadian census cycles. The 2021 ON-Marg dimensions were linked to hospital admissions at the dissemination-area level using PCCF+ postal code linkage.[39] Each dimension reflects increasing marginalization with higher values and was categorized into Ontario quintiles (1 = least, 5 = most marginalized) for analysis.

Outcomes

Our primary outcome was a composite of ICU admission and in-hospital mortality, capturing the spectrum of severe CAP. A composite outcome was used because analyzing each event separately would reduce statistical power, particularly in subgroups with low mortality, and could obscure clinically meaningful associations with early clinical deterioration. In older adults, ICU admission is additionally influenced by the competing risk of in-hospital mortality, which limits the interpretability of ICU admission as an isolated endpoint in this age group.

Secondary analyses examined ICU admission and in-hospital mortality as separate outcomes. We further described other clinical outcomes including hospital length of stay (LOS), and 7- and 30-day readmission to any medical or intensive care service at a participating GEMINI hospital following discharge from the index episode of care. An episode of care is defined as all contiguous inpatient hospitalizations to any medical or intensive care service, including hospital transfers. Index hospitalizations ending in death were excluded from the denominator in the readmission calculation, and hospitalizations were not counted as readmissions if they were coded as elective.

We also described processes of care for CAP management, including use of invasive and noninvasive mechanical ventilation and pleural drainage methods based on Canadian Classification of Health Interventions (CCI) intervention codes developed by CIHI (Supporting Information S1: Table 2), radiology, lab, and antibiotic medication orders.[41]

Statistical analysis

Baseline patient- and neighborhood-level characteristics, processes of care, and clinical outcomes were reported by age groups (18–39, 40–64, and 65+ years), sex, and neighborhood-level ON-Marg quintiles (1–5).

To evaluate whether the associations between baseline patient sociodemographic and clinical characteristics and the composite outcome were consistent across age groups, we estimated adjusted odds ratios (ORs) using generalized linear mixed models (GLMMs) in each age stratum separately. Covariates included age, sex, mLAPS, CCI,[34] and the four ON-Marg dimensions, selected *a priori* based on clinical relevance and a prespecified directed acyclic graph (Supporting Information S1: Figure 1). GLMMs used a binomial distribution with a logit link and included hospital-level random intercepts. For patients with more than one eligible admission during the study period, a single admission was randomly selected to avoid within-patient correlation in the regression models. For each model, analyses were restricted to admissions with complete data for the outcome and covariates.

Age and mLAPS were continuous, while sex, number of comorbidities, and the four ON-Marg dimensions were categorical. Adjusted associations from these models were expressed as mutually adjusted ORs with 95% confidence intervals, each interpreted in the context of the causal framework outlined in Supporting Information S1: Figure 1. ORs for continuous variables are per unit, except for age, which is per 10 years. ON-Marg quintiles were modeled as continuous, with the odds ratio representing the change in odds per one-quintile increase. This linear-trend specification reflects the ordinal structure of the quintiles, which are derived from underlying continuous principal component analysis scores. Missing data were excluded, and missingness is reported in Table 1. For our sensitivity analysis, we repeated the models after excluding encounters with aspiration pneumonia (J69.0), which is difficult to distinguish from CAP. In an additional sensitivity analysis, we assessed whether the associations between baseline characteristics and the composite outcome were stable over the study period. Because the cohort spanned the COVID-19 pandemic, we classified each admission into one of three eras based on admission date: prepandemic (April 2015–February 2020), pandemic (March 2020–March 2022), and later pandemic (April 2022–March 2024). We then conducted three complementary analyses on the full cohort, including the same clinical and sociodemographic characteristics as predictors as in the primary analyses. First, we refit the full model with era added as a categorical covariate and compared the resulting estimates with the primary model to assess whether era acted as a confounder. Second, we refit the full model separately within each era to examine whether the magnitude and direction of each association were consistent across periods. Third, to formally test for effect modification by era, we compared the era-adjusted model with a model that additionally included interactions between era and every predictor (age, sex, mLAPS, CCI, and the four ON-Marg dimensions) using a likelihood ratio test (18 degrees of freedom).

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TABLE 1 Cohort baseline characteristics.

Variables	Overall, *N* = 84,356	Age category			Sex	
		18–39, *N* = 3777	40–64, *N* = 16,120	65 and above, *N* = 64,459	Female, *N* = 38,971	Male[a] *N* = 45,385
Age						
Median (Q1, Q3)	78.00 (65.00, 87.00)	31.00 (26.00, 36.00)	57.00 (51.00, 61.00)	82.00 (75.00, 89.00)	79.00 (67.00, 88.00)	77.00 (65.00, 86.00)
Sex						
Female	38,971 (46%)	1,705 (45%)	6,891 (43%)	30,375 (47%)		
Male[a]	45,385 (54%)	2,072 (55%)	9,229 (57%)	34,084 (53%)		
mLAPS						
Median (Q1, Q3)	21.00 (12.00, 33.00)	15.00 (6.00, 25.00)	20.00 (10.00, 29.00)	23.00 (13.00, 33.00)	21.00 (11.00, 32.00)	22.00 (12.00, 33.00)
Missing	5329	374	1352	3603	2446	2883
CCI at admission						
0	35,988 (43%)	2980 (79%)	8274 (51%)	24,734 (38%)	16,984 (44%)	19,004 (42%)
1	19,041 (23%)	406 (11%)	3594 (22%)	15,041 (23%)	8581 (22%)	10,460 (23%)
2	14,694 (17%)	255 (6.8%)	2123 (13%)	12,316 (19%)	6943 (18%)	7751 (17%)
3 +	14,633 (17%)	136 (3.6%)	2129 (13%)	12,368 (19%)	6463 (17%)	8170 (18%)
Most responsible diagnosis						
Aspiration Pneumonia	14,346 (17%)	589 (16%)	1861 (12%)	11,896 (18%)	5467 (14%)	8879 (20%)
CAP	50,430 (60%)	3020 (80%)	10,433 (65%)	36,977 (57%)	24,833 (64%)	25,597 (56%)
COPD	18,747 (22%)	34 (0.9%)	3354 (21%)	15,359 (24%)	8448 (22%)	10,299 (23%)
COVID-19	431 (0.5%)	36 (1.0%)	211 (1.3%)	184 (0.3%)	125 (0.3%)	306 (0.7%)
HIV	402 (0.5%)	98 (2.6%)	261 (1.6%)	43 (<0.1%)	98 (0.3%)	304 (0.7%)
Top 10 Charlson comorbidities						
Diabetes with complications	17,328 (21%)	106 (2.8%)	2206 (14%)	15,016 (23%)	7011 (18%)	10,317 (23%)
Chronic pulmonary disease	16,026 (19%)	285 (7.5%)	3057 (19%)	12,684 (20%)	7768 (20%)	8258 (18%)
Congestive heart failure	12,120 (14%)	77 (2.0%)	1110 (6.9%)	10,933 (17%)	6209 (16%)	5911 (13%)
Diabetes without complications	11,275 (13%)	180 (4.8%)	2099 (13%)	8996 (14%)	4925 (13%)	6350 (14%)
Cancer (any Malignancy)	10,227 (12%)	203 (5.4%)	2267 (14%)	7757 (12%)	4091 (10%)	6136 (14%)
Dementia	5364 (6.4%)	suppressed ($n < 6$)	83 (0.5%)	5280 (8.2%)	2595 (6.7%)	2769 (6.1%)
Metastatic solid tumor	3343 (4.0%)	52 (1.4%)	875 (5.4%)	2416 (3.7%)	1385 (3.6%)	1958 (4.3%)
Renal disease	2634 (3.1%)	67 (1.8%)	433 (2.7%)	2134 (3.3%)	1050 (2.7%)	1584 (3.5%)
Myocardial Infarction	1171 (1.4%)	suppressed ($n < 6$)	124 (0.8%)	1045 (1.6%)	514 (1.3%)	657 (1.4%)
Mild Liver Disease	602 (0.7%)	32 (0.8%)	278 (1.7%)	292 (0.5%)	205 (0.5%)	397 (0.9%)

Abbreviations: CAP, community-acquired pneumonia; CCI, Charlson Comorbidity Index; COPD, chronic obstructive pulmonary disease; HIV, human immunodeficiency virus; mLAPS, modified laboratory-based acute physiology.

[a]This category includes "Male" = 45,382 and "Other" = 3.

To evaluate the extent to which sociodemographic factors improve model fit beyond clinical characteristics, we compared nested model specifications within each age stratum. Specifically, the full model (clinical + sociodemographic) was compared with two reduced models using nested likelihood ratio tests (LRTs): a clinical-only model and a sociodemographic-only model. These comparisons, performed separately in each age group, allowed us to assess both the contribution of sociodemographic factors to the outcome and the consistency of these associations across age strata. Analyses were conducted using R Version 4.4.1 (R Foundation for Statistical Computing Platform, 2024), with GLMMs fit using the *lme4* package.[42]

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RESULTS

Baseline patient characteristics

The total cohort included 84,356 admissions to 26 hospitals from April 2015 to March 2024 (Supporting Information S1: Figure 2), which are summarized by age and sex in Table 1. The median age of the cohort was 78 (interquartile range [IQR]: 65–87) years, and the proportion of females was similar across age strata (46% overall). Median mLAPS for the cohort was 21 (IQR: 12–33), which increased with age. Similarly, the proportion of patients with a CCI of 3 or greater rose from 3.6% in the 18–39 stratum to 19% in patients ≥65 years, while 79% of those aged 18–39 had no comorbidities (CCI = 0).

A substantial portion of the cohort resided in areas characterized by higher levels of marginalization (Figure 1). For instance, 27% lived in neighborhoods with the lowest socioeconomic position (Q5 for material resources), and 39% were in areas with the highest neighborhood-level housing marginalization (Q5 for households and dwellings). Furthermore, in the youngest age strata (18–39 years), 41% resided in areas with the highest concentration of racialized and newcomer populations and 29% in areas with the lowest unemployment, with both proportions higher than in older age strata.

Clinical outcomes

Clinical outcomes are described by age and sex in Table 2. Overall, 19% of adults hospitalized with CAP experienced the composite outcome of ICU admission or in-hospital mortality. This outcome occurred in 21% of males and 18% of females, with proportions ranging from 17% to 20% across age strata. When examined separately, in-hospital mortality was 1.9% in patients aged 18–39 years, 5.0% in those aged 40–64 years, and 13% in those ≥65 years, showing a rise with age. In contrast, ICU admission occurred in 17% of those aged 18–64 years and 10% of adults aged ≥65 years, showing a decrease with age. Mechanical ventilation followed a similar age pattern, occurring more frequently among younger patients (Supporting Information S1: Table 3). For readmissions, 5.5% of

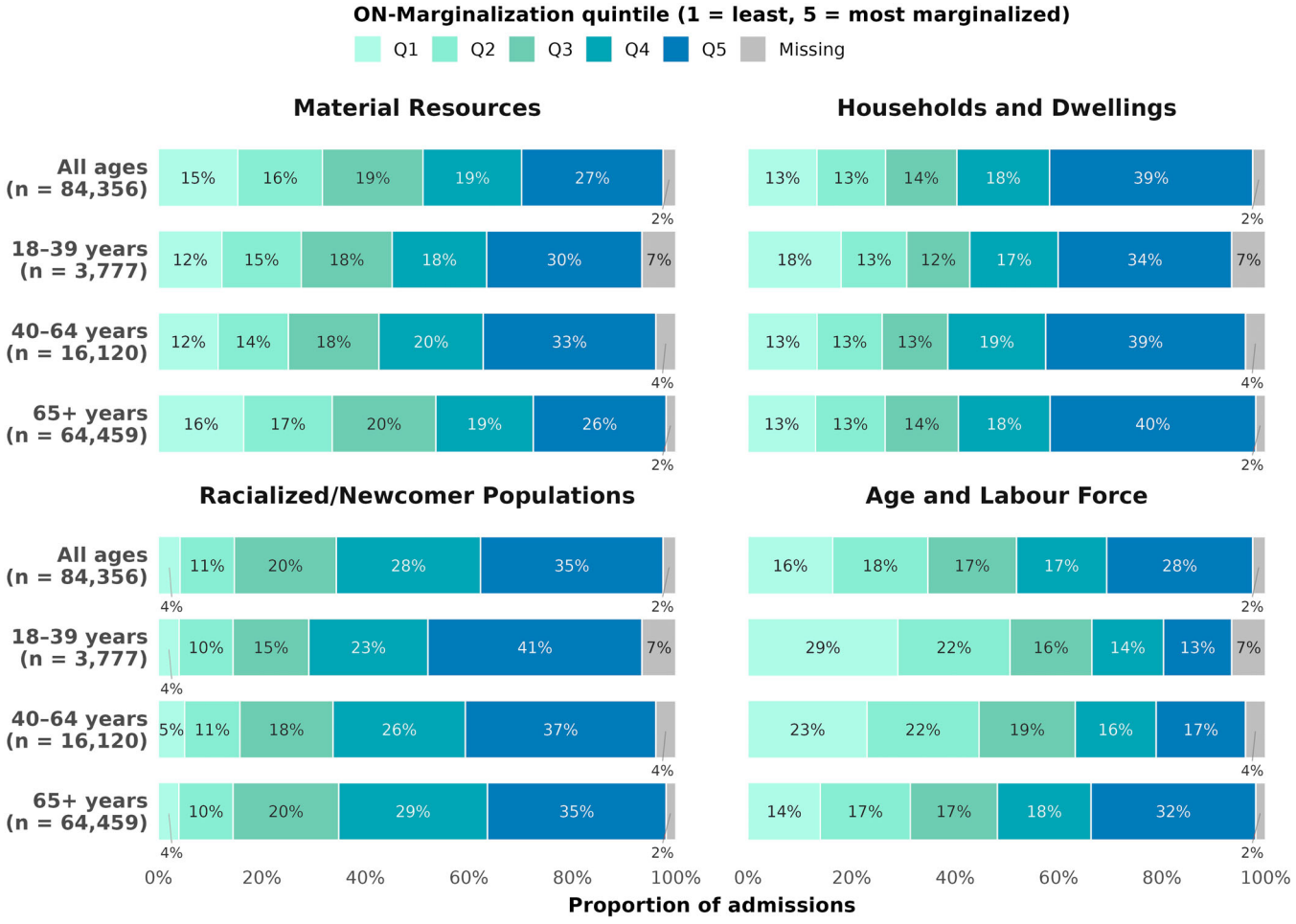


FIGURE 1 Distribution of Ontario Marginalization Index (ON-Marg) quintiles by age stratum and dimension. Each panel shows one of the four ON-Marg dimensions. Horizontal bars within each panel display the percentage of admissions in each ON-Marg quintile, stratified by age. Sample sizes (*n*) reflect the number of admissions in each stratum. Admissions with missing ON-Marg data are shown in gray. Higher quintiles indicate greater marginalization (Q1, least marginalized; Q5, most marginalized).

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TABLE 2 Clinical outcomes of CAP hospitalizations in 26 Ontario hospitals between 2015 to 2024.

Outcomes	Overall, *N* = 84,356	Age category			Sex	
		18–39, *N* = 3,777	40–64, *N* = 16,120	65 and above, *N* = 64,459	Female, *N* = 38,971	Male, *N* = 45,385
Composite outcome[a]	16,428 (19%)	659 (17%)	3084 (19%)	12,685 (20%)	6845 (18%)	9583 (21%)
ICU admission	9855 (12%)	639 (17%)	2789 (17%)	6427 (10.0%)	3963 (10%)	5892 (13%)
In-hospital mortality	9425 (11%)	70 (1.9%)	798 (5.0%)	8557 (13%)	4036 (10%)	5389 (12%)
Length of stay (days)						
Median (Q1, Q3)	5.31 (2.80, 9.74)	3.33 (1.71, 6.52)	4.40 (2.33, 8.11)	5.68 (3.05, 10.31)	5.38 (2.87, 9.63)	5.24 (2.75, 9.85)
7-day readmission[b]	*N* = 72,356	*N* = 3554	*N* = 14,682	*N* = 54,120	*N* = 33,810	*N* = 38,546
7-day readmission	3967 (5.5%)	171 (4.8%)	816 (5.6%)	2,980 (5.5%)	1,765 (5.2%)	2,202 (5.7%)
30-day readmission[b]	*N* = 71,646	*N* = 3508	*N* = 14,540	*N* = 53,598	*N* = 33,472	*N* = 38,174
30-day readmission	10,815 (15%)	412 (12%)	2155 (15%)	8248 (15%)	4748 (14%)	6067 (16%)

Abbreviations: CAP, Community-acquired pneumonia; ICU, intensive care unit.

[a]Composite of ICU admission and in-hospital mortality.

[b]Hospitalizations ending in death were excluded from the denominator in the readmission calculation, and hospitalizations were not counted as readmissions if they were coded as elective.

patients were readmitted within 7 days and 15% within 30 days. Seven-day rates were within 1 percentage point across age and sex strata, while 30-day rates were similar across sex and the older age strata but lower among patients aged 18–39 (12% vs. 15% overall) (Table 2). Results describing processes of care for the clinical management of CAP by age strata are presented in Supporting Information S1: eResults.

Association of baseline characteristics with primary outcome

The cumulative incidence of ICU admission was highest in patients under 65, especially for males, while in-hospital mortality increased with age for both sexes, ultimately exceeding the rate of ICU admission for patients aged ≥65 years (Supporting Information S1: Figure 3).

After adjusting for baseline clinical and sociodemographic characteristics, sex was strongly associated with the composite outcome across all age strata (Figure 2). Compared with female patients, the adjusted odds of the composite outcome among males were higher in all three age strata, and the point estimates were smaller in successively older strata (18–39 years: adjusted odds ratio [aOR]: 1.48, 95% confidence intervAL [CI]: 1.21–1.81; 40–64 years: aOR: 1.36, 95% CI: 1.24–1.49; and ≥65 years: aOR: 1.25, 95% CI: 1.19–1.31). These associations were similar when ICU admission and in-hospital mortality were examined separately (Supporting Information S1: Table 4).

Age was significantly associated with the composite outcome only among patients aged ≥65 years, but not for the younger age strata ($p < .001$). In this older age strata, each 10-year increase in age was associated with 9% higher odds of ICU admission or in-hospital mortality (aOR: 1.09; 95% CI: 1.07–1.12). When examined separately, increasing age was associated with substantially lower odds of ICU admission (36% lower odds per 10 years; aOR: 0.64; 95% CI: 0.62–0.66) but markedly higher odds of in-hospital mortality (50% higher odds per 10 years; aOR: 1.50; 95% CI: 1.45–1.54).

Residing in areas with a higher proportion of racialized or newcomer populations was associated with lower adjusted odds of the composite outcome for all patients (18–39 years: 0.89, 95% CI: 0.81–0.98; 40–64 years: aOR: 0.89, 95% CI: 0.85–0.93; and 65+ years: 0.96, 95% CI: 0.94–0.98). When ICU admission and in-hospital mortality were examined separately, this association was driven primarily by lower odds of ICU admission with a higher neighborhood-level racialized/newcomer population. Other ON-Marg dimensions were not significantly associated with the composite or individual outcomes.

mLAPS and CCI at admission were consistently associated with the composite outcome across all age strata. Each five-point increase in mLAPS was associated with a 10%–13% increase in the odds of ICU admission or in-hospital mortality, with similar effect sizes across age strata. Having two or more comorbidities (vs. none) was associated with higher odds of the composite outcome in patients aged 18–39 years (aOR: 1.68, 95% CI: 1.24–2.28), aged 40–64 years (aOR: 1.29, 95% CI: 1.16–1.44) and aged ≥65 years (aOR: 1.54, 95% CI: 1.47–1.62). When in-hospital mortality was examined separately, among patients with two or more comorbidities (vs. none), the adjusted odds of in-hospital mortality were higher across all age strata, with larger point estimates in younger patients: aOR: 3.98 (95% CI: 2.24–7.07) in those aged 18–39 years, 2.19 (95% CI: 1.84–2.60) in those aged 40–64 years, and 1.51 (95% CI: 1.43–1.60) in those aged ≥65 years.

The LRTs (Supporting Information S1: Table 5) showed that both sociodemographic and clinical variables were significant contributors to the composite outcome across all age strata. For every age stratum, the full model, which included both sets of variables, provided a

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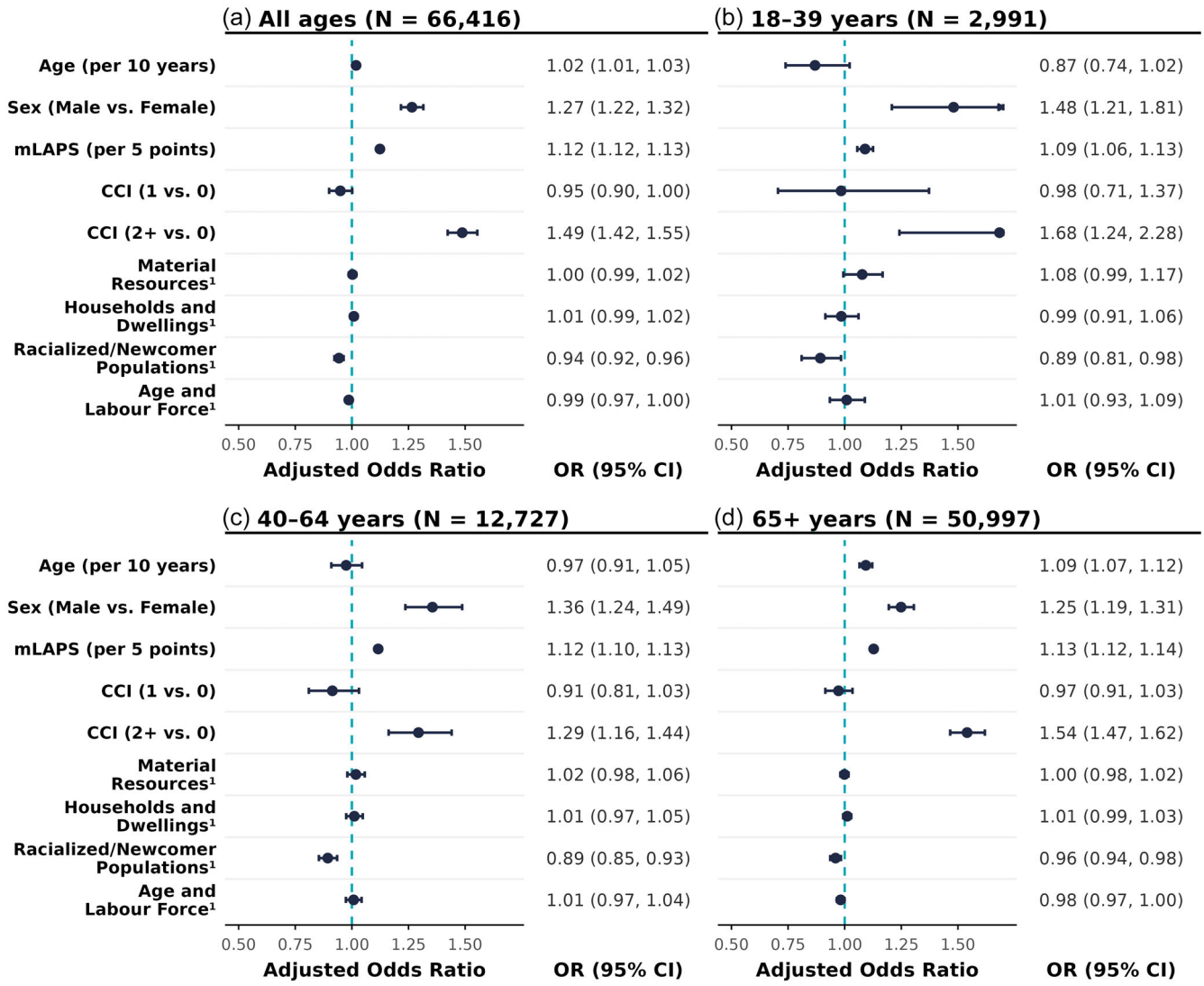


FIGURE 2 Adjusted odds ratios for the composite outcome (intensive care unit [ICU] admission or in-hospital mortality) among adults hospitalized with community-acquired pneumonia, stratified by age group. Points represent adjusted odds ratios and horizontal lines denote 95% confidence intervals from binomial generalized linear mixed models with a random intercept for hospital. The vertical dashed line indicates OR = 1.0 (no association). Panel sample sizes reflect unique patients after random selection of one admission per patient. 1 Ontario Marginalization Index (ON-Marg) quintiles modeled as continuous (1 = least marginalized, 5 = most marginalized). The OR represents the change in odds per one-quintile increase in marginalization. Higher quintiles reflect greater area-level deprivation: reduced material resources, less stable household and dwelling conditions, higher concentrations of racialized and newcomer populations, and neighborhoods with older age profiles and lower labor force participation. CI, confidence interval; CCI, Charlson Comorbidity Index; mLAPS, modified Laboratory-based Acute Physiology Score; OR, odds ratio.

significantly better fit than models that contained only sociodemographic or only clinical factors ($p < .001$ for all comparisons). However, the magnitude of improvement was consistently greater when adding clinical variables to the sociodemographic model (LRT $\chi^2 = 41.24–1682$) compared with adding sociodemographic variables to the clinical model (LRT $\chi^2 = 25.73–145.12$).

In the sensitivity analysis excluding patients with a primary diagnosis of aspiration pneumonia, these findings remained consistent (Supporting Information: Figure 4). In an additional sensitivity analysis accounting for the pandemic era in the full cohort, the associations between baseline clinical and sociodemographic characteristics and the composite outcome were robust (Supporting Information S1: Table 6). Adjusting for the pandemic era did not change the estimates from our primary analyses, and the adjusted odds ratios for male sex and for residence in areas with higher concentrations of racialized and newcomer populations were directionally consistent across the pre-pandemic, pandemic, and later-pandemic periods. Independent of these characteristics, admission during the pandemic period was associated with 33% higher odds of ICU admission or in-hospital death relative to the pre-pandemic period (aOR: 1.33, 95% CI: 1.26–1.40); these increased odds returned to baseline in the later-pandemic period (aOR: 0.96, 95% CI: 0.90–1.02). Although a likelihood ratio test

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comparing models with and without era-by-predictor interactions was significant ($\chi^2 = 116.4$, df = 18, $p < .001$), the adjusted odds ratios for the associations central to our findings were directionally consistent across all three eras.

DISCUSSION

In this multicenter, retrospective cohort study of 84,356 CAP hospitalizations, we observed age-specific associations between baseline sociodemographic characteristics and the composite outcome of ICU admission or in-hospital mortality. Male sex was consistently associated with higher risk across age groups, although the estimated association was smaller in older strata. Increasing age was associated with higher odds of the composite outcome only among patients aged 65 years and older, consistent with prior evidence identifying older age as a strong predictor of severe CAP outcomes.[10,19,43,44] Residence in areas with higher concentrations of racialized and newcomer populations was associated with a lower risk of the composite outcome. Model comparisons demonstrated that clinical characteristics, particularly greater mLAPS and greater comorbidity burden (CCI ≥ 2), explained most of the variation in the composite outcome.

Hospitalized males with CAP experienced worse outcomes than females across age groups after adjustment for illness severity, comorbidity burden, and neighborhood-level marginalization. This aligns with prior studies reporting longer hospitalization, a more complex course of CAP, higher rates of ICU admission, and greater mortality among men.[19,45–48] The literature is not entirely consistent, however: some studies have reported worse outcomes among women, although these generally assessed endpoints different from our composite of ICU admission or in-hospital mortality. For example, one found longer lengths of stay, slower clinical improvement, and a 15% higher risk of 28-day death among women,[49] and another reported higher adjusted mortality in ICU for females compared with males.[50] Such sex-based differences are thought to have a biological basis: estrogen confers anti-inflammatory and immunomodulatory effects, whereas testosterone has been linked to immunosuppression.[51,52] Consistent with this, our analysis showed that the estimated association between male sex and the composite outcome was largest in the youngest stratum and smaller in older strata. One explanation for this effect may be that premenopausal women benefit from higher estrogen levels, but this protective effect wanes after menopause, narrowing sex-related differences in later life.[53]

Although these biological mechanisms are likely important, gender-related and behavioral factors may also contribute to men's higher odds of the composite outcome. Men tend to present with greater illness severity than women at admission[48]; consistent with this, men in our cohort had higher mLAPS at admission, yet the male association persisted after adjustment for mLAPS, indicating that severity at presentation does not fully explain the disparity. Men are also more likely to smoke and consume alcohol and to have occupational and behavioral exposures that shift the distribution of causative pathogens, with organisms such as *Legionella* reported more frequently in men.[48,54] We could not measure these behavioral or microbiological factors, nor differences in care such as ICU triage thresholds that may influence a composite outcome incorporating ICU admission. Therefore, residual confounding from these unmeasured factors cannot be excluded.

ICU admission, one component of the composite outcome, showed a distinct age pattern. Among hospitalized patients, the proportion admitted to the ICU did not increase with age and was in fact lowest among the oldest patients, consistent with prior data.[55] The greater use of mechanical ventilation among younger patients, a primary indication for intensive care, may contribute to their higher rate of ICU admission.

Our cohort skewed toward being from more marginalized neighborhoods, which is consistent with prior evidence that higher neighborhood-level marginalization is associated with higher incidence of CAP.[17,56] Prior studies have reported mixed findings regarding neighborhood socioeconomic position and clinical outcomes, with some showing higher risk of severe outcomes among residents of higher marginalization areas[17,56] and others reporting no clear associations.[57,58] We observed a significant association only for the racialized and newcomer population ON-Marg dimension, with lower risk of ICU admission or in-hospital mortality among CAP inpatients in all age strata. This finding may reflect the "healthy migrant effect," whereby recent immigrants are often healthier than native-born populations in Canada, the United States, and Western Europe,[59] paralleling the Hispanic paradox reported in US pneumonia cohorts.[60] It may also reflect limitations in our data: comorbidities may be underestimated in newcomer populations, for whom less prior health information is typically available, and because the racialized and newcomer dimension is measured at the neighborhood level while the CCI is measured at the individual level, our adjustment may not fully capture individual-level differences in health status. The remaining ON-Marg dimensions showed no independent associations with the composite outcome, which may reflect several factors. Material and housing deprivation likely act on CAP outcomes largely through comorbidity and illness severity at presentation, which we captured with the CCI and mLAPS, so little independent association remained after adjustment. Similarly, the age and labor force dimension partly reflects neighborhood age structure, which overlaps with the individual-level age already adjusted for in our age-stratified models. As with the racialized and newcomer dimension, these are area-based measures, and ecological indices are relatively insensitive to individual material circumstances.[61] In a comparable universal-coverage setting, neighborhood socioeconomic position showed little association with in-hospital CAP outcomes.[57]

Key strengths of this study include its large, population-based design across 26 academic and community hospitals serving a diverse Canadian population, and a universal health care system that minimizes insurance-related barriers to hospital and ICU access, enhancing the generalizability of these findings. The GEMINI database is uniquely rich, capturing granular, encounter-level data, which provided a comprehensive description of management and clinical outcomes across a large, representative population of hospitalized patients with CAP.

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Several limitations should be considered when interpreting these findings. First, Canadian hospitals do not routinely collect self-identified race, ethnicity, or other key sociodemographic information at the patient level. As a result, we relied on neighborhood-level data derived from the ON-Marg index, which may not fully capture individual-level variation. Second, while the GEMINI data set includes a large and diverse set of hospitals across Ontario, it is possible that some patients were readmitted to hospitals outside the GEMINI network, leading to potential underestimation of readmission events. However, given the distribution of hospitals within the GEMINI network and hospital catchment areas, we believe this occurrence to be infrequent. Third, data availability varied by hospital and over time. For example, not all sites consistently submitted complete medication, imaging, or lab order data for the full study period. Analyses of these outcomes were therefore limited to timeframes and sites with complete data. Fourth, the study period spanned the COVID-19 pandemic, raising the possibility that temporal changes in case mix, hospital capacity, and care delivery influenced our findings. In a sensitivity analysis, the associations central to our findings were consistent across pandemic eras. The higher odds of severe outcomes observed during the acute pandemic period likely reflect the inclusion of COVID-19 pneumonia in our cohort together with pandemic-related strain on hospital and intensive care resources, rather than a change in the relationships between patient characteristics and the composite outcome. Finally, age-stratified analyses resulted in relatively smaller sample sizes among younger adults (18–39 years). This may limit the precision of estimates in this stratum, although it reflects real-world patterns, as CAP disproportionately affects older adults.

Overall, our study showed that in addition to clinical factors such as disease severity and comorbidity burden, baseline sociodemographic factors such as age, sex, and neighborhood-level marginalization contributed to ICU admission or in-hospital mortality for patients admitted with CAP. These findings highlight that both clinical and sociodemographic baseline characteristics affect adverse outcomes in patients hospitalized with CAP, including ICU admission and in-hospital mortality, and should be considered in early risk stratification to improve prognosis and outcomes for these patients.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Datasets from GEMINI are not permitted to be shared openly given they contain potentially sensitive patient information, based on the data governance policies of the GEMINI research network and its research ethics board-approved study protocols. Data can be accessed in the secure GEMINI research environment. Information about data access can be obtained at www.geminimedicine.ca, by contacting GEMINI. Research@unityhealth.to, or upon request to the corresponding author.

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